

MSDS OF

GLYPHOSATE-ISOPROPYLAMMONIUM 48% W/V SL

SECTION 1 - CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

Tradenames and synonyms: GLYPHOSATE-ISOPROPYLAMMONIUM 48% W/V SL

MANUFACTURER: SICHUAN LESHAN FUHUATONGDA AGRO-CHEMICAL TECHNOLOGY CO., LTD

SECTION 2 - COMPOSITION, INFORMATION ON INGREDIENTS

Components

Material	CAS Number	% (w/v)
Glyphosate	1071-83-6	48%
Other inert ingredients:		52%

SECTION 3 - HAZARDS IDENTIFICATION

Appearance and odor: Orange red liquid

Warning statements: Keep out of the reach of children.

Warning!

Causes substantial but temporary eye injury.

Harmful if swallowed or inhaled.

Do not get in eyes or on clothing.

Avoid breathing vapor or spray mist.

Potential health effects:

Eye: May cause pain, redness and tearing based on toxicity studies.

Skin: No more than slightly toxic no more than slightly irritating based on toxicity studies.

Ingestion: No more than slightly toxic based on toxicity studies. No significant adverse health effects are expected to develop if only small amounts (less than a mouthful) are swallowed. Ingestion of similar formulations has been reported to produce gastrointestinal discomfort with irritation of the mouth, nausea, vomiting and diarrhea. Oral ingestion of large quantities of one similar product has been reported to result in hypotension and lung edema.

Inhalation: No more than slightly toxic if inhaled based on toxicity studies.

Carcinogenicity information: None of the components present in this material at concentrations equal to or greater than 0.1% are listed by IARC, NTP, OSHA or ACGIH as a carcinogen.

SECTION 4 - FIRST AID MEASURES

If inhaled: remove to fresh air. If not breathing, give artificial respiration. If breathing is difficult, give oxygen. Call a physician.

If on skin: Flush skin with water after contact. Wash contaminated clothing before

reuse.

If in eye: In case of contact, immediately flush eyes with plenty of water for at least 15 minutes. Call a physician.

If swallowed: do not induce vomiting. Immediately give 2 glasses of water. Never give anything by mouth to an unconscious person.

SECTION 5 - FIRE FIGHTING MEASURES

Flash point: > 93°C

Extinguishing media: Water spray, foam, dry chemical, CO₂

Fire fighting instructions: Wear self-contained breathing apparatus. Wear full protective equipment.

SECTION 6 - ACCIDENTAL RELEASE MEASURES

Personal precaution

Wear appropriate protective gear for the situation. See personal protection in section 8.

Methods for cleanup and disposal

Small spills-For a spill less than gallon on floor or other impervious surface, soak up with towels or other absorbent material and discard in trash. Clean the spill area with soap and water and rinse the area thoroughly.

Large liquid spills- On the floor or other impervious surface should be contained or diked and then absorbed with attapulgate, bentonite or other absorbent clays. Collect the contaminated absorbent, place in metal drum and dispose of in accordance with the instructions provided under Disposal. Thoroughly scrub floor other impervious surface with a strong industrial detergent and rinse with water.

Large spills that soak into the ground should be dug up. Place in metal drums and disposed of in accordance with instructions provided under Disposal. Contact appropriate state agency when considering a land spreading disposal option.

Leaking containers should be separated from non-leakers and either the container or its content transferred to a drum or other non-leaking container and disposed of in accordance with instructions provided under disposal. Any recovered spilled liquid should be similarly collected and disposed of.

SECTION 7 - HANDLING AND STORAGE

Handling

Do not get in eyes, on skin or clothing. Avoid breathing vapors or mist. Wash thoroughly with soap and water after handling. Do not store or consume food, drink or tobacco in areas where they may become contaminated with this material. Wash clothing after use.

Storage

Do not contaminate water, foodstuff, feed or seed by storage or disposal.

SECTION 8 - EXPOSURE CONTROLS, PERSONAL PROTECTION

Personal protective equipment

Eye/face protecting: Wear safety glasses or coverall chemical splash goggles.

Respirators: Where there is potential for airborne exposures in excess of applicable limits, wear NIOSH approved respiratory protection if there is potential for exposure to airborne dusts, mists or vapors..

Protective clothing: Where there is potential for skin contact have available and wear as appropriate impervious gloves, apron, pants, and jacket.

SECTION 9 - PHYSICAL AND CHEMICAL PROPERTIES

Physical data

Color: Orange red liquid

Odor: little odor

Form: Liquid.

pH: 4.0-8.5

Specific gravity: 1.19

SECTION 10 - STABILITY AND REACTIVITY

Chemical stability: Stable at normal temperatures and storage conditions.

Incompatibility with other materials

Spray solutions of this product should be mixed, stored, or applied using only stainless steel, aluminum, fiberglass, plastic or plastic-lined containers.

Do not mix, store or apply this product or spray solutions of the product in galvanized or unlined steel (except stainless steel) containers or spray tanks. This product or spray solutions of this product react with such containers and tanks to produce hydrogen gas which may form a highly combustible gas mixture. This gas mixture could flash or explode, causing serious personal injury, if ignited by open flame, spark, welder's torch, lighted cigarette or other ignition source.

Polymerization

Do not occur. This product can react with caustic basic materials to liberate heat. This is not a polymerization but rather a chemical neutralization in an acid base reaction.

SECTION 11 - TOXICOLOGICAL INFORMATION

Animal data

ingle exposure (acute) studies indicate:

Oral- practically non-toxic Rat LD50 > 5000 mg/kg

Dermal- practically non-toxic Rabbit LD50 > 5000 mg/kg

Inhalation- slightly toxic Rat 4hr LC50 – 2.6 mg/l

Eye irritation- Rabbits 24 hr –slight to moderate irritation

Eye irritation was evident at day 14 but cleared by day 21 after exposure in 1 animal.

Skin irritation-Rabbits 4 hr- essentially non-irritating.

Slight erythema (redness) clearing in all animals within 24 hours.

No skin allergy was observed in guinea pigs following repeated skin exposure.

SECTION 12 - ECOLOGICAL INFORMATION

Ecotoxicological information

Glyphosate herbicide has been shown to be slightly to moderately toxic in aquatic studies. Glyphosate herbicide has been shown to be practically non-toxic to avian species following subacute dietary exposure.

Rainbow Trout 96-hr LC50 22mg/l (static)

Rainbow trout 96-hr LC50 8.2 mg/l (dynamic)

Daphnia magna 48-hr LC50 37 mg/l (aeration)

Daphnia magna 48-hr LC50 24 mg/l (w/o aeration)

Bluegill sunfish 96-hr LC50 5.8 mg/l (dynamic)

Bluegill sunfish 96-hr LC50 14 mg/l (static)

Gammarus pseudolimnaeus 48-hr EC50 42 mg/l

Fathead minnow 96-hr LC50 9.4 mg/l

Channel catfish 96-hr LC50 16 mg/l

Chinook salmon 96-hr LC50 20 mg/l

Coho salmon 96-hr LC50 22 mg/l

Algae S. Capricornutum 72-hr EC50

Bobwhite quail 8-day LC50 > 6300 ppm

Mallard duck 8-day LC50 > 6300 ppm

SECTION 13 - DISPOSAL CONSIDERATIONS

Waste Disposal Methods:

Waste resulting from the use of this product that can not be used or chemically reprocessed should be disposed of in a landfill approved for pesticide disposal or in accordance with applicable federal, state or local procedures. Emptied container retains vapor and product residue. Observe all label safeguards until container is destroyed.

Container handling and disposal:

Plastic Bottles and Non-Returnable Plastic Drums: Do not reuse container. Triple rinse container. Then puncture and dispose of in a sanitary landfill, or by incineration, or, if allowed by state and local authorities, by burning. If burned, stay out of smoke.

Returnable / refillable containers: close all openings which have been opened during use and replace all caps.

SECTION 14 - TRANSPORT INFORMATION

Shipping information

DOT

Not regulated.

SECTION 15 - REGULATORY INFORMATION

U.S. Federal regulations

TSCA Inventory Status: All components in compliance with TSCA inventory requirements.

SICHUAN LESHAN FUHUA TONGDA AGRO-CHEMICAL TECHNOLOGY CO.LTD.

QIAOGOU TOWN, WUTONGQIAO DISTRICT, 614000 CHINA

SECTION 16 - ADDITIONAL INFORMATION

The MSDS summarized our best knowledge of the health and safety information of the product and how to safely handle and use the product in the work[place. The date in this Material Safety Data Sheet relates only to the specific material designated herein and are based upon data believed to be correct.

ORIGINAL CONTROLLED